

Jinhyung Lee, Ph.D.

Postdoctoral Research Associate, Department of Biomedical Engineering, University of Connecticut

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Google Scholar: 286 citations, h-index 6, i10-index 5 (July 2026) | U.S. Permanent Resident (No Visa Sponsorship Required)

EDUCATION**University of Connecticut, Storrs, CT, USA** 09/2019–05/2024*Ph.D., Biomedical Engineering*

Research Advisor: Prof. Yupeng Chen

Dissertation: “DNA-Inspired Nanopieces of Drug Delivery in Biomedical Applications”

University of California, San Diego, La Jolla, CA, USA 09/2015–06/2017*M.S., Bioengineering*

Research Advisor: Prof. Yi Chen

Thesis: “Self-Assembly DNA Nanogel with Embedded Functional Proteins for Nanomedicine Platform”

Konkuk University, Seoul, Republic of Korea 03/2009–02/2015*B.S., Convergence Biotechnology, College of Bioscience and Biotechnology*

Mandatory military service: Republic of Korea Air Force, 2010–2012

PROFESSIONAL EXPERIENCE**University of Connecticut, Storrs, CT, USA** 09/2024–present*Postdoctoral Research Associate, Department of Biomedical Engineering***Korea Institute of Science and Technology (KIST), Seoul, Republic of Korea** 09/2017–12/2018*Researcher, Brain Science Institute***RESEARCH EXPERIENCE****University of Connecticut, Storrs, CT, USA** 09/2024–present*Research Advisor: Prof. Yupeng Chen*

- Leading development of mRNA therapeutics for osteoarthritis using the Janus base nanopiece (JBNp) platform; Principal Investigator, ORS/Bethel Fellowship Grant
- Engineered computation-aided rod-shaped Janus base nanoparticles for solid tumor targeting; co-first author, *J. Control. Release* (2025)
- Led nanoparticle co-delivery of doxorubicin and siRNA to overcome multidrug resistance; co-corresponding author, *Materials Today Advances* (2026)

University of Connecticut, Storrs, CT, USA 09/2019–05/2024*Research Advisor: Prof. Yupeng Chen*

- Developed the Janus base nanopiece (JBNp) platform with enhanced endosomal escape; first author, *PNAS* (2021)
- Established siRNA, mRNA, and CRISPR-Cas9 delivery protocols with demonstrated in vitro and in vivo efficacy

Korea Institute of Science and Technology (KIST), Seoul, Republic of Korea 09/2017–12/2018*Researcher, Brain Science Institute*

- Developed GPCR-based FRET biosensors for real-time neurotransmitter detection; first-named inventor, Korean Patent KR 10-2176538

University of California, San Diego, La Jolla, CA, USA 01/2016–06/2017*Research Advisor: Prof. Yi Chen*

- Developed a self-assembling DNA nanogel platform with embedded functional proteins for intracellular delivery

TEACHING EXPERIENCE**Teaching Assistant, Foundations of Biomechanics (BENG 226, graduate level)** 04/2017–06/2017*Department of Bioengineering, University of California, San Diego*

- Led weekly discussion sections, prepared supplementary instructional materials, and graded problem sets and examinations

PEER-REVIEWED PUBLICATIONS

7 publications | 4 as first or corresponding author | † Equal contribution | * Corresponding author

First-Author Publications

1. †Lee J, †Zhang W, Nguyen D, Zhou L, Amengual J, Zhai J, Cote T, Landolina M, Ahmadi E, Sands I, Mishra N, Yu H, Nieh M, *Wang K, *Li Y, *Chen Y. “Computation-aided design of rod-shaped nanoparticles for tumoral targeting.” *Journal of Controlled Release*. 2025;387:114169. [IF 11.5]

2. **Lee J**, Sands I, Zhang W, Zhou L, *Chen Y. “DNA-inspired nanomaterials for enhanced endosomal escape.” *Proceedings of the National Academy of Sciences USA*. 2021;118(19):e2104511118. [IF 12.7]
3. †Yau A, †**Lee J**, *Chen Y. “Nanomaterials for protein delivery in anticancer applications.” *Pharmaceutics*. 2021;13(2):155. [Highly Cited Paper, Web of Science]

Corresponding-Author Publications

4. †Zhai J, †Surian A, Cote T, Zhang W, Dang BT, Wang Y, Chen Q, Hoshino K, ***Lee J**, *Chen Y. “Enhanced endosomal escape for nanoparticle-enabled co-delivery of doxorubicin and siRNA to overcome multidrug resistance.” *Materials Today Advances*. 2026;30:100845. [IF 8.0]

Co-Authored Publications

5. Zhou L, Zhang W, **Lee J**, Kuhn L, *Chen Y. “Controlled self-assembly of DNA-mimicking nanotubes to form a layer-by-layer scaffold for homeostatic tissue constructs.” *ACS Applied Materials & Interfaces*. 2021;13(43):51321–51332. [Cover Article]
6. Sun M, **Lee J**, Chen Y, Hoshino K. “Studies of nanoparticle delivery with in vitro bio-engineered microtissues.” *Bioactive Materials*. 2020;5(4):924–937. [IF 8.7]
7. Sands I, **Lee J**, Zhang W, *Chen Y. “RNA delivery via DNA-inspired Janus base nanotubes for extracellular matrix penetration.” *MRS Advances*. 2020;5(16):815–823.

Book Chapters

1. Sands I, **Lee J**, *Chen Y. “Nanotechnology for DNA and RNA delivery.” *Nanomedicine: Technologies and Applications*, Chapter 8. Elsevier, 2023.
2. **Lee J**, *Chen Y. “Nucleic acid delivery systems for gene therapy.” *Gene Delivery: Methods and Applications*, Chapter 2. Nova Science Publishers, 2019.

GRANTS AND FUNDING

ORS/Bethel Fellowship Grant, Orthopaedic Research Society 10/2026–09/2027
 Role: Principal Investigator | Award: \$20,000 | Cosponsor: Orland Bethel Family Musculoskeletal Research Center

Co-Investigators: Prof. Yupeng Chen (University of Connecticut), Dr. Chuan-Ju Liu (Yale School of Medicine)

PATENTS

1. Chen Y, **Lee J**, Zhang W. “Therapeutic nanomaterials.” U.S. Patent No. 12,214,043 B2, issued February 4, 2025 (appl. 17/514,390). Assignee: University of Connecticut.
2. Chen Y, Zhang W, **Lee J**. “Nanomaterial delivery vehicle and method of use thereof.” U.S. Patent Application Publication US 2025/0302762 A1, published October 2, 2025 (appl. 18/852,925; PCT WO 2023/150684 A2). Pending.
3. **Lee J**, Seong J, Kim H, Choo H. “FRET-based serotonin 2A biosensor.” Korean Patent KR 10-2176538, granted 2020.

PRESENTATIONS

4 Invited Talks | 2 Podium Presentations | 20 Conference Presentations

Invited Talks

1. **Lee J**. “DNA-inspired Janus base nanoparticles for drug delivery in biomedical applications.” Chonnam National University Medical School Seminar, Republic of Korea, 2026.
2. **Lee J**. “Computation-aided design of rod-shaped Janus base nanotubes for enhanced tumoral targeting.” Biomaterials and Biofabrication Conference, USA, 2025.
3. **Lee J**. “DNA-inspired nanopieces of drug delivery in biomedical applications.” College of Medicine, Kyung Hee University Seminar, Republic of Korea, 2024.
4. **Lee J**, Chen Y. “DNA-inspired nanomaterials for small molecule delivery to hard-to-penetrate solid tumors.” Next-Generation Nanomedicine, USA, 2023 (short talk).

Podium Presentations

1. **Lee J**, Cote T, Zhai J, Dang BT, Chung EJ, Landolina M, Chen Y. “Rod-shaped DNA-inspired Janus base nanoparticles enable in vivo kidney CRISPR editing.” BMES Annual Meeting, Orlando, FL, October 2026 (accepted).
2. **Lee J**, Cote T, Davidson N, Zhai J, Zhang W, Morgan K, Liu C, Laurencin C, Chen Y. “Janus base nanopieces delivered mRNA therapeutics for osteoarthritis treatment.” Orthopaedic Research Society Annual Meeting, 2024.

AWARDS AND RECOGNITION

- ORS/Bethel Fellowship Grant, Orthopaedic Research Society (\$20,000), 2026
- Predoctoral Fellowship, University of Connecticut, 2023
- RNATx Young Investigator Travel Award, UMass Chan Medical School, 2023
- Graduate Travel Awards (4 competitive awards), University of Connecticut, 2020–2022
- Sung-jo Merit Scholarship (highest merit tier) and Shin-jo Merit Scholarship, Konkuk University, 2014–2015

LEADERSHIP, MENTORING, AND SERVICE

Graduate and Undergraduate Students Mentored: 7 total

- Trained students in Janus base nanomaterial synthesis, nanoparticle characterization (dynamic light scattering, zeta potential, fluorescence and confocal microscopy), and RNA delivery protocols
- Co-authored 4 conference presentations with mentees at BMES and SCCUR

Scholarly Leadership and Service

- Reviewer, ASSAY and Drug Development Technologies (SAGE Publications), 2026
- Member: Orthopaedic Research Society; American College of Rheumatology; Biomedical Engineering Society